

The Socioeconomic Effects of Forced Displacement: Evidence from the Tennessee Valley Authority*

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Abstract

We examine the socioeconomic effects of forced migration by focusing on individuals who were displaced by the Tennessee Valley Authority (TVA) dam projects in the 1930s. We use data from the relocation program associated with the TVA and link them to US Census data. We compare individuals who were impacted by the dam-induced flooding to those who will be displaced by different TVA dam projects at a later date. We find evidence that individuals who were displaced by the dam projects are more likely to participate in the labor force, marked by an inflow of labor into unskilled occupations and an outflow from skilled occupations. We examine racial disparities in outcomes and find that, after the relocation, Black men are more likely to be in these unskilled occupations compared to White men.

Keywords: Displacement, Labor, TVA

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1 Introduction

Forced displacement resulting from wars, natural disasters, and human rights violations has become increasingly common. The United Nations estimated the number of forced displaced individuals globally reached roughly 120 million at the end of 2025 [United Nations High Commissioner for Refugees \(2026\)](#). Understanding the consequences of forced displacement is therefore becoming increasingly important. Its severe impacts can likely persist long after the incident. Previous studies have shown evidence of negative impacts of relocation driven by natural disasters on labor market outcomes [Mueller and Quisumbing \(2009\)](#) and financial resources [Chen \(2020\)](#). Other studies have examined the impacts of the Great Migration on African Americans in the United States (U.S.) [Tolnay \(2003\)](#). There is little evidence on short- and long-term impacts of forced displacement induced by government development projects, although this practice of eminent domain is very common in the U.S.

This paper investigates the effects of forced migration by leveraging a natural experiment from the 1930's - the Tennessee Valley Authority (TVA) dam projects. The TVA created dams in seven states in the southeastern region of the U.S. as part of the New Deal, displacing thousands of people. We use data on the displaced individuals, which we have matched to U.S. Census records using a probabilistic matching algorithm, to examine the effects of forced displacement on a range of medium-run socioeconomic outcomes. Development of these reservoirs was an effort to combat the lack of economic opportunity in the region, especially after the downturn of the Great Depression. However, the success of this New Deal program on the people it displaced is not widely understood.

The Tennessee Valley Authority Act of 1933 marked the first time a government agency had power to govern the development of an entire region. Not only did the act improve infrastructure and rural life, it drastically improved the region's standard of living within a relatively brief period [Walker \(1998\)](#). "The T.V.A. has relocated over 12,000 farm families, moved entire villages, helped whole communities plan a better future, and withal it has demonstrated that social planning can win the enthusiastic support of the

people at the muzzle end of it. So the T.V.A. has set a pattern for social planning for a generation" Carr (1948). However, historical research suggests the TVA perpetuated segregation and discrimination against Black individuals Jones (2016); Alderman and Brown (2010). For instance, using TVA's relocation records and oral histories, Walker (1998) argues the Authority's benefits were not fairly distributed. The agency maintained a policy of institutionalized segregation and discrimination against Black people which also prevented them from deriving direct benefits from TVA's employment and educational programs. Walker (1998) also states that the beneficial programs provided by the TVA favored prosperous landowners over poorer landowners and tenants of both races. Although relocation workers attested to abiding by TVA's removal worker's code of ethics, many assumed African Americans did not have a sense of community, hence only using the word "community" in documentation towards White families. This suggests TVA workers were less sympathetic about removing Black families from their communities than for White families. A chief criticism of the dam construction project is the lack of a systematic follow up of the removed families to their new locations Brown (1951). The author states, "If this was done, a better basis would be available for judging the success and failures of family removal and relocation in the Tennessee Valley."

This paper provides that missing systematic follow-up. Using a difference-in-differences design, we find that forced relocation by the TVA increased displaced men's labor force participation by 3.3 percentage points (10.2%), though displacement had no effect on the extensive margin of employment. Displaced men were also more likely to work in unskilled occupations and less likely to be in skilled occupations, a pattern driven primarily by an influx into non-farm unskilled roles rather than agriculture. These results suggest that while forced relocation may boost labor market entry, these gains are accompanied by a shift toward lower-skilled work rather than upward occupational mobility.

We further find differential impacts by race: Black men were 13.6% more likely than White men to be in the labor force and 53.8% more likely to be in unskilled occupations following displacement. Using a race-matched sample, we show that the contraction in skilled employment among displaced Black men was nearly triple the effect size seen

among White men, a disparity that is statistically significant and consistent with differential treatment in the post-displacement labor market. We also find that displaced male renters, rather than owners, drove the improvements in labor force participation and employment, while displaced owners instead saw declines in employment and skilled occupations. These patterns do not appear to be explained by selection into treatment or by relocation to systematically better counties, suggesting instead that they reflect the disruptive effects of displacement itself.

This study contributes to different areas of the literature. First, it adds to the body of work that examines the effects of displacement through the lens of the Great Migration in the United States. Although the identification strategy in these studies is affected by the fact that African Americans made a decision to migrate north for better opportunities, harsh living conditions have been argued to leave many with little choice but to relocate. Existing evidence shows that relocation had positive impacts: increased earnings of Black migrants [Collins and Wanamaker \(2014\)](#) and schooling for migrant children, which decreased the White-Black education gap [Baran et al. \(2024\)](#). However, [Dernoncourt \(2022\)](#) finds that destination cities responded to an influx of Black migrants by changing city characteristics, such as increasing policing, thus reducing the benefits of migration. This response to an influx of Black migrants also led to an increase in incarceration rates for Black individuals [Eriksson \(2019\)](#). We add to these studies by examining how Black and White people respond similarly and differently to a more short-distance forced relocation.

We also contribute to the literature on the impact of displacement after natural disasters. For example, previous studies have shown that Hurricane Katrina had small impacts on victims' wage earnings and income, with labor supply and earnings returning to pre-disaster levels for those who relocated back after the hurricane [Deryugina et al. \(2018\)](#); [Vigdor \(2007\)](#). However, these natural disasters can cause financial hardship, especially for poorer countries and counties with greater income disparity [Johar et al. \(2022\)](#); [Cavallo et al. \(2022\)](#); [Lim et al. \(2017\)](#). The [National Research Council \(2006\)](#) states: "...more recent research suggests that such [positive rebounding from natural disasters] findings paint an overly simplified and perhaps overly optimistic picture of post-disaster recovery."

In addition, we contribute to the literature on the impact of displacement caused by government development programs. For instance, [Chyn \(2018\)](#) and [Chetty et al. \(2016\)](#) examine the impact of distinct policies within the Moving to Opportunity literature that led to the demolition of public housing and provided housing vouchers to move people from high-to-low-poverty neighborhoods in the 1990s. They show that this form of displacement improves the economic outcomes of children who moved. These studies raise the question of whether providing an opportunity to move, rather than forcing relocation, is better for citizens in that area. The results of [Chyn \(2018\)](#) suggest that forcing relocation leads to beneficial longer-term outcomes for children, perhaps because short-term constraints prevented such moves when they were not forced. However, development projects from the 1996 Atlanta and 2008 Beijing Olympic games pushed migrants further out into poorer areas, resulting in negative long-term impacts [Richardson \(2022\)](#); [Shin and Li \(2013\)](#). The literature on the interstate highway system has documented its negative impacts, especially on the African American community, by destroying communities and creating inner-city ghettos [Karas \(2015\)](#). However, in India, the creation and expansion of highways improved the socioeconomic well-being of people living in its proximity (although they were not displaced) [Sengupta et al. \(2007\)](#).

We add to the ‘New Deal’ literature by examining the effects of its establishment. Many New Deal programs were shown to offset the repercussions of the Great Depression [Fishback et al. \(2007\)](#). The authors saw that an increase in relief spending from these New Deal programs led to a decline in infant mortality, suicide rates, and deaths from infectious diseases. The majority of grant and loan programs were geared towards the poor and unemployed and raised income, increased property values near new public housing projects, while also reducing property crime rates [Fishback and Kachanovskaya \(2016\)](#); [Fishback and Kollmann \(2015\)](#); [Fishback et al. \(2010\)](#). However, certain programs also led to a reduction in employment and labor force participation, especially for women [Blundell et al. \(2016\)](#); [Sundstrom \(2001\)](#).

The existing economic literature on the TVA has not been able to examine individual responses to the program. The TVA reservoirs led to a temporary increase in malaria

rates, later reduced by TVA’s anti-malaria programs [Kitchens \(2013\)](#). Its electrification had no significant effect on the productivity of farming or manufacturing [Kitchens \(2014\)](#). However, [Kline and Moretti \(2014\)](#) find that the growth rate of manufacturing employment was significantly larger in TVA counties than non-TVA counties; hence, TVA sped the industrialization of the Tennessee Valley. The authors focus on the economy of TVA counties compared to non-TVA counties, while our paper will focus on the people who were impacted by the development from the TVA.

This project contributes to the literature on forced relocation more broadly. Similar papers in this literature look at the impacts of large dam projects on local economic outcomes. Dams constructed in India and Africa had negative impacts, such as increased poverty and lower productivity croplands, on those near the construction versus those living downstream of the dam, who benefited [Duflo and Pande \(2007\)](#); [Strobl and Strobl \(2011\)](#). Other papers focus on government administered forced relocation programs in Europe following war [Hornung \(2014\)](#); [Becker et al. \(2020\)](#); [Sarvimäki et al. \(2022\)](#), and catastrophic natural disasters [Nakamura et al. \(2022\)](#). Importantly, these migrations featured racially and ethnically homogeneous groups. A recent review of the forced migration literature notes that little work examines how “contextual variables” like race influence economic responses to forced migration [Becker and Ferrara \(2019\)](#). In our setting, we examine racial heterogeneity especially as the TVA relocation program occurred during the era of Jim Crow in the southern United States. We provide the first causal evidence of the effects of government sponsored forced relocation on individual labor market outcomes, and document disparate impacts for Black and White individuals.

Overall, our findings show that forced relocation from the Tennessee Valley Authority’s dam construction project improved labor market outcomes. The defining aspect of this project is that these events, like extreme weather events, are forced and citizens do not have a choice of whether to stay or leave their homes.

The paper is organized as follows. Section 2 provides background information on the creation of the Tennessee Valley Authority and its relocation program. Section 3 describes the data sources used to identify forced relocation effects. It also contains an explana-

tion of our linking process to historical census. Our empirical approach is discussed in Section 4, followed by our conceptual framework. The results are presented in Section 5 along with a robustness check and mechanisms. We conclude in Section 6.

2 Background

2.1 Historical Background: Tennessee Valley Authority and the Great Depression

The 1930s Great Depression was one of the worst economic depressions in history. During the time, many Americans placed blame on President Herbert Hoover for steep unemployment and massive bank failures. His successor, President Franklin D. Roosevelt, began the ‘New Deal’ recovery program to stimulate economic recovery. The Tennessee Valley Authority (TVA) was established through the Tennessee Valley Authority Act of 1933 as part of the New Deal. The duties of the organization included improving the navigability of the Tennessee River; providing flood control through reforestation of marginal lands in the Tennessee Valley watershed; developing agriculture, commerce and industry in the valley; and operating the hydroelectric dams¹. The Act also gave TVA the power of eminent domain, allowing them to acquire land along the Tennessee River for developmental reasons. These hydroelectric dams, shown in Figure 1, were constructed to expand the Tennessee River, moving east to west, and runs through the seven states that TVA oversees: Alabama, Georgia, Kentucky, Mississippi, North Carolina, Tennessee, and Virginia. Construction of these dams caused numerous towns to be submerged underwater, forcing residents to relocate. On average, construction of each dam took about 3 years to complete between 1933 and 1953.

2.2 Tennessee Valley Authority’s Relocation Program

The TVA provided a relocation program to assist families displaced by dams move to a new location. TVA sent interviewers to each household that would be affected. These interviewers filled out a relocation application with the family that contained detailed information about their house, land, and demographics for all individuals living there.

¹Transcript of the [National Archives and Records Administration](#) (1933)

This information was then used by the TVA to determine suitable dwellings for families to relocate to. Though the TVA did not provide compensation for relocation, it provided its resources to find new homes for sale, appraised them, and relocated families to farms that best suited their needs [Satterfield \(1937\)](#). To our knowledge, a rubric that details how and in which order new dwellings were chosen does not exist. However, it has been documented that families were relocated to a new home and land that was similar to their previous home and land [McCarthy \(1949\)](#); [Satterfield \(1937\)](#). Although most families were tenant farmers, families and landlords who owned their homes and land were offered fair market prices [McCarthy \(1949\)](#). Relocation was provided for both the living and deceased², and an estimated 15,000 families and 30,000 graves were relocated across all seven states.

To our knowledge, there were not clear documented rules determining the TVA's decision to flood particular areas. However, [Howard \(1936\)](#) explains the steps towards dam construction the TVA took in order to implement the program as efficiently as possible. In the twelve step plan towards the project completion, fully interpreted into a list in the Appendix Figure [A1](#), the majority of the steps were focused on the employees hired to construct the dams. These included understanding the labor supply of the region as well as providing resources for those employees such as a camp to house them and resources to educate them on-and off-duty. Other steps focused on cooperation between local governments and the TVA as the region developed and the project progressed. One of the last steps pertained to the relocation of families. As mentioned, detailed facts about families who were in the path of the dam construction were obtained; these details were used by social scientists to analyze those who were being displaced and to determine relocation strategy. [Howard \(1936\)](#) also notes that an analysis determining the extent to which [relocated] families have successfully integrated themselves into a new life, without either cultural or economic loss, will be important to document the success of the program. To our knowledge, a full analysis as such has yet to happen³.

²The Tennessee Valley Authority obliged by the family's wishes to relocate their loved one's grave.

³TVA issued a descriptive report on the resettlement of the population of each major dam constructed in the 1930s. We will provide causal inference about resettlement using economic models.

3 Data

3.1 Information on Relocated Families

We obtained information on relocated families in Alabama and Tennessee from the “U.S., Tennessee Valley, Family Removal and Population Readjustment Case Files, 1934-1953” dataset via ancestry.com. This dataset contains pdf files of original relocation applications which provides rich information about each relocated family in order for the TVA to decide where to move them (Figures II.I and II.II). Housing details such as ownership of property, dwelling characteristics, and farming style of crops were described at the household level. Employment information such as occupation, income, and highest level of education were provided for all members of the family living in the house at the time. For each family member, demographics such as gender, race, birth year, relation to the head of the household, along with their full name were provided as well. To our knowledge, we are the first to transcribe and digitize pdf files of TVA’s relocation applications for the states of Alabama and Tennessee, which constitutes over 7800 relocated families.

3.2 Census Data

After obtaining information on which families were relocated, we linked those families to the 1930s census by matching on name and other defining characteristics such as birth year, gender, and residing county, to gather our outcome variables. The census also allowed us to gather outcome variables for non-relocated families who were not in the path of the dam construction which constitutes our control group(s). These include a farming indicator, an ownership indicator, monthly rent payments, labor force participation, employment, and occupation score. We obtain data prior to dam construction from the 1900, 1910, 1920, and 1930 censuses, with post-construction data coming from the 1940 census.

We use record linking methods to link the data on relocated families to the 1930 census data using a probabilistic matching algorithm that matches individuals on name, birth

year, location, and gender. We require an exact match on county and gender, and allow for flexible matching on both name and age. However, we require that individuals be born in the same decade and they have the same first and last initial. We are able to link a share of the relocated families data to the 1930 census. Of the linked relocatees, a share were linked to the 1940 census. Lastly, a share of all relocatees were linked across previous censuses. The population of linked relocated families contains more males, White individuals, farmers, have higher income and literacy rate on average, and less likely to rent their property compared to those who did not link across earlier censuses. Our final sample consists of families and of men in the state of Alabama who have both 1930 and 1940 census information and linked information for at least one of the 1900, 1910, or 1920 census.

4 Empirical Approach

4.1 Empirical Strategy

To understand impacts of a forced relocation, we compute a simple difference-in-difference model that observes outcomes of those who were forced to relocate, the treated group, compared to those who were similar but not-yet forced to relocate due to timing of the dams construction, the control group, after the TVA Act was enacted in 1933. These individual labor outcomes consist of employment status, labor force status, and the switch to unskilled or skilled occupations [Ferrie \(1997\)](#). We look solely at men because married women were not consistently asked about employment when censuses were conducted. Therefore, women and children are excluded from individual employment outcomes.

Specifically, our treatment group contains individuals displaced by the Wheeler (1933) and Guntersville (1935) dams in Alabama and the Norris dam (1933) in Tennessee. The control group contains individuals displaced by the Cherokee (*late 1940*), Fort Loudoun (*late 1940*), Ocoee No. 3 (1941), Douglas (1942), Boone (1950) and Fort Patrick (1951) dams in Tennessee and the Nottely dam (1941) in Georgia. The placement of the dams and the locations of individuals in our treated and control groups are shown in [Figures 1 and 3](#) respectively.

We estimate a difference-in-difference to determine the effects of a forced relocation in equation (1):

$$Y_{it} = \beta_0 + \beta_1 Black_i + \beta_2 Treated_i + \beta_3 Post_t + \beta_4 Treated_i * Post_t + \tau_t + \epsilon_{it} \quad (1)$$

where Y_{it} represents the socioeconomic outcomes for an individual male i in census year $t \in \{1930, 1940\}$. $Black_i$ is a binary variable that equals 1 if the race of the male is Black and 0 if race is White⁴. $Treated_i$ indicates whether the men were displaced by the dam construction. $Post_t$ represents the decennial census period, 1940, after the relocation time period. The interaction between $Treated_i$ and $Post_t$ indicates a displaced man in the post period of 1940. β_4 is our coefficient of interest, representing the socioeconomic effects of being displaced by the TVA's dam construction projects. Our error term is represented by ϵ . All regressions report county-clusted standard errors. For results by race, we use equation (2):

$$Y_{it} = \beta_0 + \beta_1 Treated_i + \beta_2 Post_t + \beta_3 Treated_i * Post_t + \tau_t + \epsilon_{it} \quad (2)$$

where we drop the $Black_i$ indicator variable and run the regression separately by race. β_3 becomes our coefficient of interest and the rest of the equation follows the description of equation (1).

4.2 Identifying Assumption

A key assumption underlying our difference-in-differences design is that treated and control counties would have followed similar trends in the absence of dam construction. Because our main analysis relies on decadal Census data, we cannot directly assess pre-treatment trends in the primary sample⁵. To provide indirect evidence for this assumption, we construct event study plots using six annual county-level proxy outcomes drawn from Fishback and Kantor (2018): population, CCC camp presence, total New Deal spend-

⁴We drop nonblack and nonwhite races due to a small amount of observations. There is also no evidence of multiple races per household in our sample during this time period.

⁵For a detailed decadal summary of the statistics, please refer to the Appendix, Table A1

ing per capita, WPA and FERA spending per capita, manufacturing employment, and farming assistance. These outcomes span a range of demographic and economic dimensions and allow us to assess whether treated and control counties were trending similarly in the years leading up to dam construction. Importantly, many of these outcomes proxy for other New Deal programs that could potentially affect the labor market outcomes of individuals impacted by dam construction.

Figure 4 presents the results. Across all six proxy outcomes, pre-treatment coefficients are small, statistically indistinguishable from zero, and show no systematic directional drift in the years prior to dam construction. This pattern holds consistently throughout the pre-period before dam construction, suggesting that treated and control counties were on broadly similar trajectories before treatment. Although these proxies do not constitute a direct test of parallel trends for our primary outcomes, the consistency of the null pre-trend findings across six independently measured series provides reassuring evidence that our identifying assumption is plausible. Additionally, there does not appear to be a significant change in any of the proxy outcomes after dam construction, which provides suggestive evidence that we are capturing the true treatment effect of forced displacement.

4.3 Conceptual Framework

To inform our empirical work, we discuss the possible channels in which our outcomes may respond to forced relocation. First, consider the setting of our study; Alabama, Tennessee, and Georgia are primarily rural states that host many farmers on their land. With the creation of the dam project, the TVA sought workers from the local region to help with construction. As shown in the Appendix, Figure A1, the steps towards implementation of the dam construction focused mainly on the employees. With an emphasis on the employees and the job availability that the dam project would bring, we expect to see an increase in employment. As the availability for employment increases, a forced relocation might also provide the opportunity of occupation switching. For example, construction jobs for the dam project is seen as an unskilled occupation, hence we foresee an influx

into unskilled occupations whereas we might see a decrease into skilled occupations.

With the TVA Act of 1933 occurring in the midst of the Segregation Era in the South, we expect to see differences in effects between Black and White relocatees. As stated above, we anticipate the dam construction project causing an increase in employment. However, this increase may have favored White men. This hypothesis is grounded in the historical documentation of the TVA, where seemingly neutral policies often led to unequal outcomes. [Walker \(1998\)](#) documents the “first come, first served” approach was frequently used to preserve community demographics, which systematically limited opportunities for Black families while favoring White families.

5 Results

5.1 Main Results

We report the effects of TVA relocation program on labor market outcomes in [Table 1](#) and graphically in [Figure 5 \(a\)](#). Our estimates indicate a 3.3 percentage point (p.p.) increase in labor force participation for displaced males compared to those who have not yet been displaced (a 10.2% increase). While displacement had no effect on the extensive margin of employment, displaced men were 6.4 p.p. more likely to work in unskilled occupations and 4.2 p.p. less likely to be in skilled occupations⁶. However, our placebo test in [Table 3](#) reveals a significant pre-existing divergence in skilled occupation between 1920 and 1930, prior to dam construction. Given this evidence of a pre-trend, we do not interpret the skilled occupation result as a clean causal estimate. We report it for completeness, but our preferred estimates of occupational effects rely on the unskilled occupation result (column 3), for which the placebo test shows no evidence of differential pre-trends. These results suggest that forced relocation increased labor market entry, but primarily through an expansion into unskilled, non-farm work rather than occupational upgrading.⁷

⁶Unskilled occupations and skilled occupations are defined using the occupation classification scheme in [Long and Ferrie \(2013\)](#), where unskilled occupations include laborers and service workers and skilled occupations include managers, craftsmen, and other professionals

⁷We decompose the unskilled occupation category into non-farm and farm-related roles—following the classification in [Long and Ferrie \(2013\)](#) determine if the shift into unskilled work is driven by a movement away from agriculture. As shown in [Table A2](#), the increase in unskilled occupations is primarily driven by an influx into non-farming roles (Column 2), compared to the small incline in unskilled farm labor (Column 3).

Table 2, and Figure 5 (b) graphically, depicts differential impacts of forced relocation by race. Overall, both Black and White men see outcomes in the same direction as our main results, which indicates an increase in labor market participation and unskilled occupations, and a decrease in skilled occupations. We run a triple interaction analysis using an indicator variable that helps identify any difference in outcomes by race. In the last column in Table 2, we report no significant difference in employment levels nor in skilled occupations for both Black and White men. This means that men, regardless of race, experienced similar treatment in these areas after relocating. However, we do show a significant difference in labor force participation and unskilled occupations. Specifically, Black men were 13.6% more likely to be in the labor force, yet 53.8% more likely to be in unskilled occupations compared White men. These results suggest differences in the impact of a forced relocation by the TVA between Black and White men.

5.2 Robustness Checks

We assess the validity of our identification strategy through a placebo test, a test of selection into treatment, and estimation on alternative matched samples. Table 3 reports a placebo test comparing 1920 and 1930 outcomes, a period entirely prior to dam construction. Consistent with parallel trends, we find no significant divergence in employment rate, labor force participation, or unskilled occupation share between eventually-treated and control counties.⁸ We do, however, find a significant pre-existing divergence in skilled occupation share of -11.8 p.p., which we address directly in our discussion of the main results above.

Table 5 regresses treatment status on each of our four outcome variables measured in 1930, controlling for race, to assess whether observable labor market characteristics in 1930 predict subsequent selection into treatment. We find no evidence that any outcome significantly predicts treatment assignment, supporting the assumption that treatment status is not driven by pre-existing differences in observable labor market characteristics.

We further estimate our main specification on a county-matched sample (Table 7),

⁸Coefficients range from -0.001 to 0.022 p.p. and are not statistically distinguishable from zero across these three outcomes.

both including all counties (Panel A) and separately excluding counties where dam construction occurred in the 1930s (Panel B) or 1940s (Panel C), to address concerns that our results are driven by any particular decade of dam construction. Our core occupational results are stable across all three panels: displaced men remain 5.5–5.8 p.p. more likely to work in unskilled occupations and 2.2–3.4 p.p. less likely to work in skilled occupations, closely tracking our main estimates.⁹ Employment rate, insignificant in our main specification, becomes marginally significant and positive in the matched sample, suggesting our main estimate is if anything conservative.

Finally, because dam construction for all treated counties occurs within a single inter-census interval, our design is a standard two-period difference-in-differences estimator. As such, our estimates are not subject to the recently-documented biases arising from staggered treatment timing (e.g., Goodman-Bacon 2021; Callaway and Sant’Anna 2021; Sun and Abraham 2021), since there is no variation in treatment timing within our estimation sample. Taken together, these results indicate that our findings are robust to alternative sample restrictions and are not subject to the biases that typically arise from staggered treatment adoption, with the exception of the skilled occupation result noted above.

5.3 Mechanisms

One potential mechanism driving the results is that individuals were relocated to places that were better at baseline before the TVA projects began. To examine whether this is the case, we calculated the average county level characteristics for each county in Alabama in the 1930 Census. We then assign the average county characteristics to each individual in our data based on their county of residence in 1930 and 1940, and estimate the difference-in-differences in equation (1). The results of this analysis are shown in Table 4. From this analysis, our small and insignificant results show that people displaced by the TVA projects moved to places that were largely similar to where they were living before displacement.

⁹Labor force participation estimates similarly remain stable across panels, ranging from 2.3–2.8 p.p.

A second potential mechanism is that Black individuals who were relocated faced discrimination. To test this, we match the individuals in our sample with the same-race individual who is the nearest neighbor on baseline demographics and labor market outcomes. Then, we estimate the effect of forced relocation on this matched sample. Table 8 depicts the differential impacts of forced relocation by race using our tightly controlled, race-matched sample. Overall, both Black and White men see labor market outcomes move in the same direction as our main findings, characterized by an expansion in labor force participation and unskilled occupational sorting alongside a contraction in skilled employment. However, the magnitude of these effects varies sharply across racial groups. For White men (Panel A), displacement by the TVA projects induces a 1.0 percentage point increase in employment and a 1.6 percentage point increase in labor force participation, though neither is statistically significant. However, they do face a statistically significant 2.7 percentage point reduction in skilled employment.

In contrast, Black men (Panel B) experience a much larger 7.8 percentage point contraction in skilled employment, which is statistically significant at the 1% level and nearly triple the effect size seen among White men. Furthermore, while White men experience a flat response in unskilled sorting (0.3 percentage points), Black men see a 3.0 percentage point increase in unskilled occupations. Crucially, as reported in the final row of Table 8, formal coefficient equality tests reveal that the racial gap in skilled occupational downgrading is the only disparity that is statistically significant. This sharp divergence provides compelling evidence of differential treatment and racial discrimination within the post-displacement labor market, as the disruptive shock of forced relocation disproportionately penalized Black workers by restricting their access to skilled roles.

A third potential mechanism driving the results is that there is differential treatment between renters and homeowners. One potential concern related to this is that individuals were selected into treatment based on their household wealth or other baseline characteristics. We test this in Table 5 by regressing treatment status on our baseline outcomes in the 1930 Census. We find no evidence that individuals were selected into treatment based on observable characteristics. In Table 6 we examine whether there are differen-

tial outcomes between renters and owners. We find positive significant effects in labor force participation, unskilled, and skilled occupations for displaced male renters compared to not yet displaced male renters. We see no effects in labor force participation, and unskilled occupations for displaced men who were owners. However, there is a significant decline in the employment rate and skilled occupations for displaced male owners compared to their not yet displaced counterparts. The last row in Table 6 shows these estimates are significantly different for renters and owners. Specifically, displaced male renters are 9.9% more likely to be employed, 8.3% more likely to be in the labor force, and yet 62.1% more likely to be in unskilled occupations and 14.5% less likely to be in skilled occupations compared to displaced male owners. Results indicate that the TVA did not favor homeowners over renters, and -in fact- renters are driving the increase in labor market outcomes.

6 Conclusion

Understanding how forced displacement can affect socioeconomic outcomes is an area of concern for researchers and policymakers. We utilize a natural experiment provided by the forced displacement induced by the TVA dam projects to estimate the causal effect of forced displacement on socioeconomic outcomes. We find evidence that forced displacement actually improved individuals labor market outcomes by increasing labor force participation. This is accompanied by a shift from skilled occupations to unskilled occupations.

When organizing displacement, the importance of racial disparities in socioeconomic outcomes and whether government programs contribute to these disparities or ameliorate them should also be of concern. Building on a large body of research showing evidence that government programs can exacerbate racial inequality, we find evidence that the TVA had disparate outcomes for Black and White individuals. Specifically, we find that while Black men had larger increases in labor force participation than White men, they also faced a large increase in unskilled occupation. This can indicate that TVA may have not mitigated the racial disparities between Black and White individuals.

In general, we believe that the TVA dam construction projects had a positive impact on those forced to relocate. To put our estimates in perspective, we use labor estimates from American Community Survey and U.S. Bureau of Labor Statistics data. Applying a 3.3 p.p labor force participation shift to today's average U.S. county workforce of 54,365 would mean instantly adding roughly 1,794 people into the active labor force. A magnitude of 6.4 p.p. applied to today's unskilled labor would transfer 1,922 jobs, while a -4.2 p.p. would be a loss in transfer of roughly 1,022 skilled jobs to the local economy. These estimates suggest an overall positive net benefit in the extensive margin of labor supply but a noticeable effect on the labor heterogeneity.

Our research points to the vital importance of understanding the impacts that relocation programs have on individuals and how those impacts vary by race. More research is needed to understand how forced migration can affect where individuals move and whether the specific location of an individual drives the disparities found in our research.

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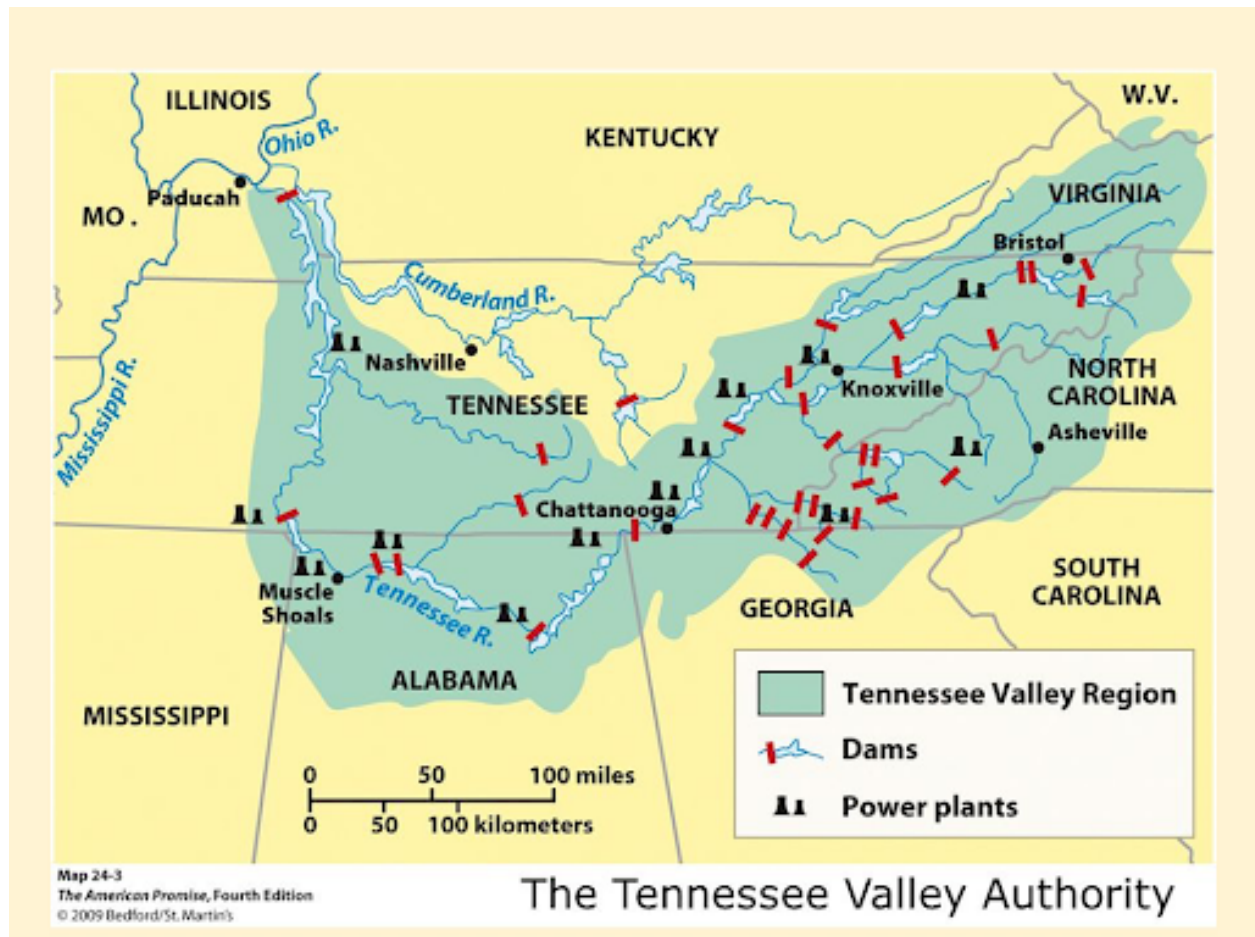
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Figures

Figure 1: Map of Tennessee Valley Authority's dam construction projects



Notes: Map is from *The American Promise, Fourth Edition*.

Figure 2: Sample Relocation Application

WPA 100
A. Tennessee Valley Authority (497)
Telephone No. 923 INFORMATION
Sex Male
Date 8/18/38 CONCERNING FUGITIVE
Social and Economic Division
Fieldwork 8/20/38
County Shelby
Civil District

1. Name J.C. Malone
2. P. O. address Athens, MS
3. Highway or road on which home is located
4. Miles from home to nearest elementary school 1 1/2 miles
5. Miles from home to nearest high school 7 1/2 miles
6. Miles from home to nearest church 1 1/2 miles
7. Chief trade or other for family
8. Average number of trips there per week Miles from home -
9. Years lived at present residence 24 In present community 24 years
10. Informative concerning houses lived in during 1934 and 1935:

Year	Address	No. of Rooms	No. of Sleeping Rooms	Age of House	Water	Electric	Telephone	Fuel	Plumbing	Other	Type of Occupancy
1934											
1935	Home	5	3	35	no	yes	yes	wood	plumbing	yes	single

11. Source of water supply: Well -
Other: Distance from nearest public water supply
12. Toilet facilities: Flush -
13. Electrical status: No. of lights 3 No. of fans 2 No. of outlets 2
14. Information concerning husband and wife:

Age	Age When Married	No. of Children	Married	County	State	Highest School Grade Reached	Condition of Health or Physical Defects
59	21	1	Married	Shelby	Tenn.	8th	Good
57	19	1	Married	"	"	8th	Good

15. No. of children born: Boys 6 Girls 5 No. who have died: Boys 2 Girls 2
16. The newspapers and magazines that the household receives regularly:
Newspapers: Louisville Courier-Journal Magazines: None
Newspapers: Nashville Tennessean

17. The approximate number of all books in the home 50
18. Denominational preference: Baptist Membership in other organizations: None
19. Amount of insurance: Life 0 Accident and Health 0
Savings 0 Other 0 Total \$200
20. Liabilities: Mortgage 0 Other debts 0 Total 0

21. Information concerning members of the family who left home to work after 1930 but later returned and now live at home:

Name	Relation	Type of Work Done	Leave taken at Work	Date left Home	Date returned Home	Married While away
<i>None</i>						

22. Information concerning deaths in the family during past ten years:

Name of Person	The Place	Date of Death	Relatives	Residence at time of death	Cause of Death	Where Buried
<i>Carl</i>	<i>Home</i>	<i>1930</i>	<i>son</i>			

23. Does the family run a car? *Yes*. If so, state the make *Ford Model A*
Age *5* How many years have they owned it *6 months*

24. Check ownership of the following personal possessions:
Radio _____ Organ _____ Sewing machine *Yes* Piano _____
Furniture _____ Other articles _____

25. The approximate value of furniture:

Room	Value	Room	Value	Room	Value
Kitchen		Living Room		Other Rooms	
Dining room		Bedrooms		Total Furniture	<i>250</i>

26. What is husband's calling or trade? *Farming*

27. Information on husband's employment during past 10 years (include farming):

Date	Kind of Work Done	Earnings per month	By Whom and Where Employed	Cause of Change in Employment
<i>None</i>				
<i>None</i>				

28. Kind of work performed

29. Information concerning husband's land (1935):

Location of Land	Tenure	Total Acres	Amount in Coddard Forests	Other	Level	Soils	Source
In Home							
In Farms	<i>None</i>	<i>12.0</i>	<i>12</i>	<i>2.7</i>	<i>71</i>	<i>1.0</i>	<i>12.0</i>
Acres	<i>None</i>						
Outside	<i>None</i>						
Acres	<i>None</i>						
Outside	<i>None</i>						
Acres	<i>None</i>						
Outside	<i>None</i>						
Acres	<i>None</i>						

30. Give location of all land other than home farm, stating whether owned or rented

31. Information concerning acres cultivated (1935): Cereals *30* Cereals *24*
Hay *10* Small grains *4* Legumes *0* Other crops *0* Total *34*

32. How many acres with trees

8

28. Information concerning type of farming and livestock (1938):

Type of Farming	No. Acres	Value	Horses	Cattle	Hogs	Sheep	Poultry
	No. Acres	No. Value	No. Value	No. Value	No. Value	No. Value	No. Value
Cattle	34	4,860	1,795	5,125	3,415		65,500

29. Type or types of farming preferred also known as 865

30. If any land is rented, give name of the owner Yana
 Is he related to this family? Yes If so, give relationship is

31. To agreement or lease with the owner written or oral?
 When was agreement begun? How does it end?

Give information concerning the terms of this agreement for year 1939:

Share of Crops Given to Owner	Cash Payment to Owner	Permitted by Owner			
		Livestock	Machinery	Fertilizer	Seed
<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>

32. If any land is rented out, give name of the tenant
 Is he related to this family? If so, give relationship

33. To farm owned by a 1934-35 action mortgage? If so, in whose name?
 In 1934 No. of acres of mortgaged land No. of acres of action owned
 No. of acres of cultivated land under mortgage No. of acres of this land
 to be flooded? If so, name, no. of acres in 1934

34. Own the of of your farm? Was one dead that will
 Was he by ?

35. Farm expenditures and receipts for 1939:

Expenditures	Amount	Receipts	Amount
Livestock and poultry	\$ <u>18</u>	Cotton	\$ <u>400</u>
Feeds	\$ <u>150</u>	Wool	\$ <u> </u>
Farm implements and machinery	\$ <u>10</u>	Hay	\$ <u>48</u>
Machinery repairs	\$ <u>10</u>	Seed	\$ <u> </u>
Farm labor	\$ <u>150</u>	Livestock	\$ <u>130</u>
Commercial fertilizers, manure	\$ <u>72</u>	Feather and poultry products	\$ <u>40</u>
Tools, plants, trees, etc.	\$ <u> </u>	Fur and milk products	\$ <u>40</u>
Taxes	\$ <u>72</u>	Forest products	\$ <u> </u>
Insurance of buildings, crops	\$ <u> </u>	Other products	\$ <u> </u>
Other (Phone repairs, etc.)	\$ <u>55</u>	Cash receipts, handicrafts	\$ <u>200</u>
Depreciation of Machine	\$ <u>30.0</u>	Other receipts	\$ <u>1,250</u>
TOTAL expenditures, 1939	\$ <u>380.0</u>	Total receipts, 1939	\$ <u>1,250</u>

36. Information concerning cash income for twelve months ending Dec. 31, 1939:

Source	Amount	Source	Amount
Wages		Children away from home	
Pensions		Childs at home	
Dividends		Industrial companies	
Government		Farm receipts	\$ <u>1,300</u>
Income		Other receipts	
		Total cash income	\$ <u>1,300</u>

37. If on relief in 1939, give estimated value of groceries, clothing, fuel, etc. (not included in cash reported) Has any relief been received during 1939 Yes If so, approximate amount

42. Family living (Products furnished by the firm and used by the family, 1944)

Products	Quantity	Value	Products	Quantity	Value
Butter, pounds	2 85	\$1.70	Peanuts, bushels	1	\$1.00
Milk, gallons	2 25	\$4.50	Garden, Acres	1/2	\$1.50
Eggs, dozen	122	\$12.20	Irish potatoes (other than		
Vegetables, bushels	122	\$12.20	in garden), bushels	2 5	\$2.50
Pork, live, dressed	100	\$10.00	Sweet potatoes (other than		
Corn, bushels	25	\$2.50	in garden), bushels	2 5	\$2.50
Wheat, bushels			Hay, pounds		
Corn, gallons			Feed, sacks	1	\$1.00
Wheat, bushels			Other products		
Apples, bushels			Total		

43. Farm machinery:

Article	Age	Value	Article	Age	Value
Wagon	3 5	\$1.50	Fertilizer distributor		
Tractor			Small tools		
Hay rake			Harrows		
Grinder			Other		
Hay baler			Total Value		
Stack rubber	10	\$1.00			
Disc harrow	10	\$1.00			

44. Has new location been definitely selected? Yes, in what county

Community	Y. or address

45. Reasons for selection: Beaumont, \$7.50

Did a real estate agent select? Yes, real estate agent

46. If no location has been selected, does the individual desire to relocate?

a. Near his present community?

b. In the same county but another vicinity?

c. Elsewhere in the Tennessee Valley?

d. In the South but outside the Tennessee Valley?

e. Elsewhere (name) _____

47. Will family buy or rent? Appropriate amount to invest

48. Size of house desired? Is electricity desired

49. Acres wooded? Crop land? Feature Woodland Total

50. Has employment been secured? If so, what? Income 177.00

51. What type of employment is desired? Farm? Part-time farm and industry

Industry? Other mercantile or professional employment

52. Special trades in which husband is skilled, such as automobile, carpentry, brick masonry, etc.

53. Give names and addresses of close relatives of this family living in this section, and give relationship to head of the family

Name	Address	Relationship

54. How did the individual respond to the interview?

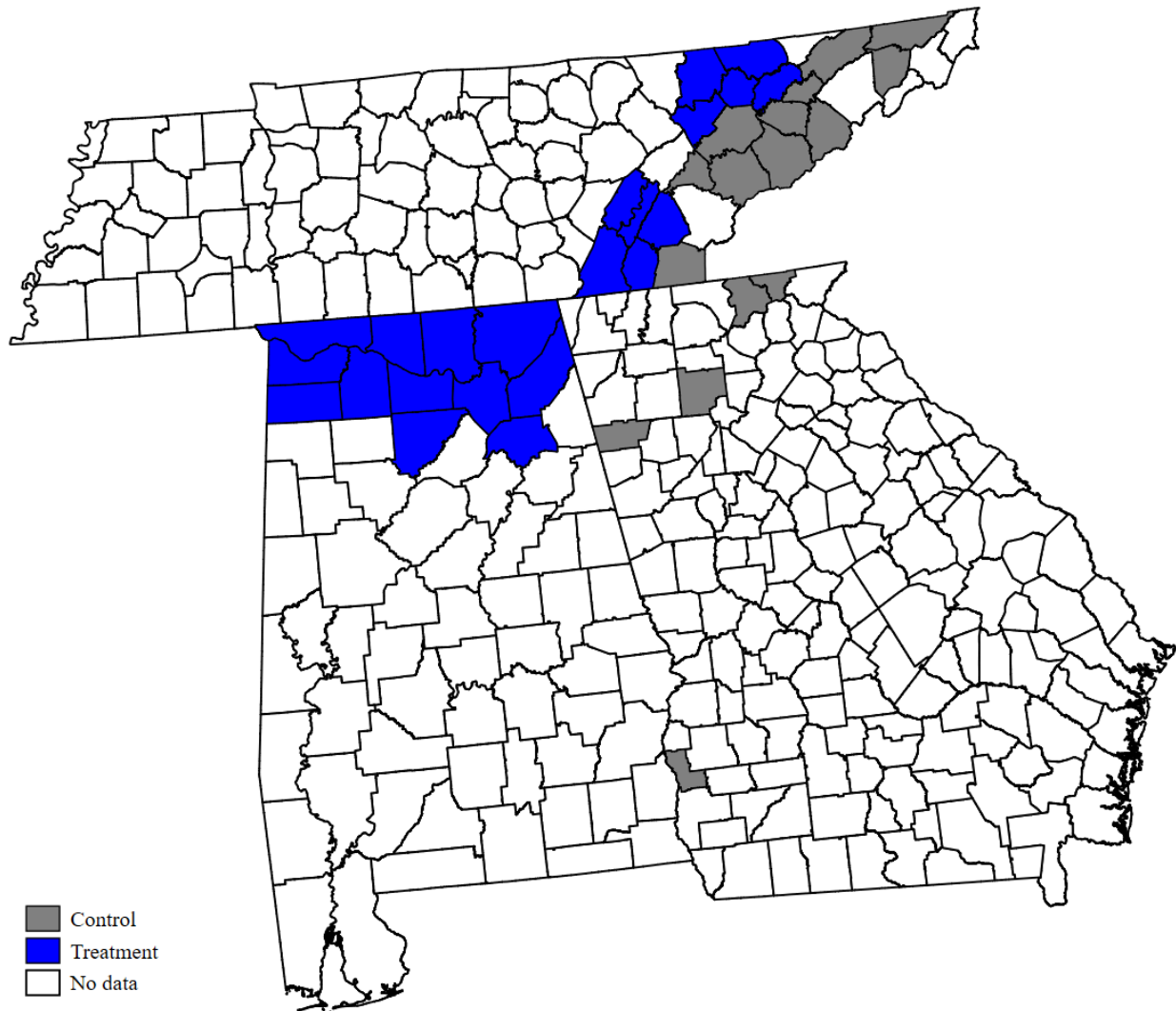
Enthusiastic	Responsive	Indifferent	Intimidated	Grudgingly Cooperator

55. What is his attitude toward the VA?

Antagonistic	Critical	Neutral	Interpreted	Active Resister

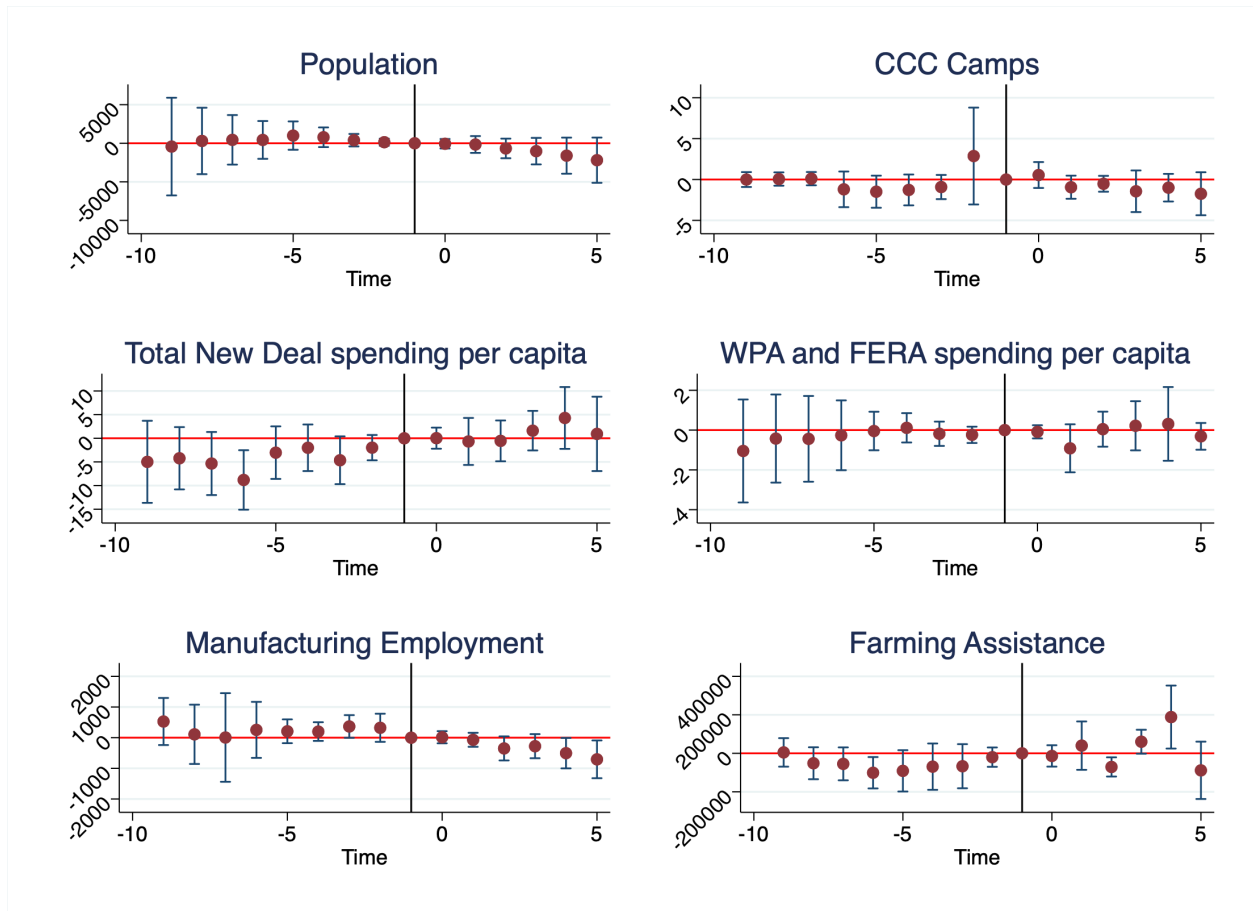
56. Write any comments regarding special problems or characteristics of this family on the back of last sheet.

Figure 3: Treatment and Control Counties, 1930s
Alabama, Georgia, and Tennessee



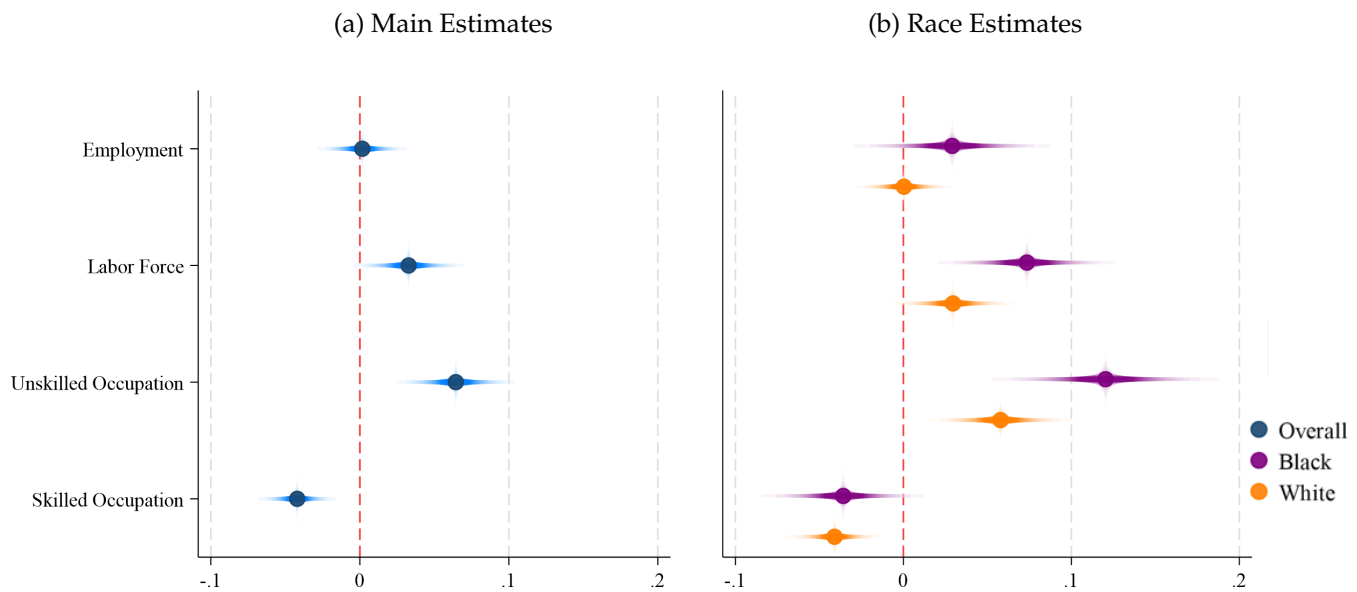
Notes: Our treatment and control groups consists of individuals living in counties that are in three states: Tennessee (top), Alabama (left), and Georgia (right). Counties consisting of treated individuals, prior to displacement, are highlighted in blue, while counties with control individuals are highlighted in gray. Counties without any treated or control individuals are not apart of our sample and are highlighted in white.

Figure 4: Parallel trends tests of annual county-level economic measures



Notes: Figures display event study estimates of the effect of dam construction on county-level proxy outcomes in the years surrounding treatment, as reported in Fishback and Kantor (2018). The sample includes 16 treated and 12 control counties. Each coefficient represents the difference between treated and control counties relative to the year prior to dam construction, which is the omitted period. The vertical line marks the year of dam construction. The red horizontal line is drawn at zero for reference. Confidence intervals are constructed at the 95 percent level and are based on standard errors clustered at the county level. These figures serve as indirect evidence for the parallel trends assumption; because the main analysis relies on Census data, annual pre-treatment trends cannot be assessed directly in the primary sample.

Figure 5: The Average Treatment Effect of a Forced Relocation on Outcomes, Main Results and by Race.



Note: This figure is a point estimate plot of both our (a) main findings and (b) main findings by race. The shaded gradient bands, spreading left and right of our solid-circled point estimates, represent a 95% confidence interval. The vertical dashed red line sits at zero on the x-axis to help visualize statistical significance. The navy blue dots represents our main findings from Table 1 and the purple and orange dots represents our race findings from Table 2, Panels A and B respectively.

Tables

Table 1: Effect of TVA relocation on socioeconomic outcomes

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
Treat*Post	0.002 (0.011)	0.033** (0.013)	0.064*** (0.015)	−0.042*** (0.010)
Observations	317,790	317,790	317,790	317,790
R-squared	0.025	0.037	0.021	0.074
Mean	0.312	0.324	0.119	0.131

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Race controls are included in each regression. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 2: Effect of TVA relocation on socioeconomic outcomes, by race

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. White</u>				
Treat*Post	0.000 (0.010)	0.030** (0.013)	0.058*** (0.015)	-0.041*** (0.010)
Observations	301,251	301,251	301,251	301,251
R-squared	0.026	0.037	0.008	0.073
Mean	0.309	0.321	0.110	0.134
<u>B. Black</u>				
Treat*Post	0.029 (0.022)	0.074*** (0.019)	0.120*** (0.025)	-0.036* (0.018)
Observations	16,454	16,454	16,454	16,454
R-squared	0.020	0.039	0.028	0.022
Mean	0.371	0.395	0.290	0.0722
Coefficients Different By Race	No	Yes	Yes	No

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A reports outcomes for White individuals in the main sample, Panel B reports outcomes for Black individuals in the main sample. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Placebo test of the effect of TVA relocation on socioeconomic outcomes

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
Treat*Post	0.022 (0.019)	0.003 (0.015)	-0.001 (0.014)	-0.118*** (0.017)
Observations	164,704	201,908	201,908	201,908
R-squared	0.047	0.068	0.036	0.099
Mean	0.314	0.311	0.108	0.109

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is a placebo indicator (1930) for being after dam construction. Employment Rate has a smaller sample size due to the sample collection of this variable in the 1920's Census. We cluster standard errors at the 1920 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: Effect of TVA relocation on baseline county characteristics in 1930

VARIABLES	(1) 1930 Average Emp. Rate	(2) 1930 Average Labor Force	(3) 1930 Average Unskilled	(4) 1930 Average Skilled
Treat*Post	−0.003 (0.002)	−0.002 (0.004)	0.001 (0.003)	0.006 (0.009)
Observations	317,790	317,790	317,790	317,790
R-squared	0.579	0.707	0.588	0.762
Mean	0.521	0.546	0.200	0.206

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Race controls are included in each regression. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5: Effect of labor market outcomes from 1930 on the probability of being in the treatment group

Variable	Treatment (Std. Err.)
Employment Rate	-0.000 (0.000)
Labor Force Participation	-0.000 (0.000)
Unskilled Occupation	-0.000 (0.000)
Skilled Occupation	0.000 (0.000)
Observations	158,973
R-squared	0.962

Note: We regress treatment status on each outcome in the year 1930. Race controls are included in each regression. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 6: Effect of TVA relocation on socioeconomic outcomes, by homeownership status

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. Renter</u>				
Treat*Post	0.007 (0.014)	0.035* (0.017)	0.087*** (0.015)	-0.054*** (0.009)
Observations	150,829	150,829	150,829	150,829
R-squared	0.035	0.052	0.025	0.076
Mean	0.298	0.314	0.131	0.123
<u>B. Owner</u>				
Treat*Post	-0.030** (0.014)	0.003 (0.013)	0.014 (0.015)	-0.036*** (0.013)
Observations	166,961	166,961	166,961	166,961
R-squared	0.017	0.025	0.019	0.074
Mean	0.327	0.335	0.108	0.139
Coefficients Different By Owner	Yes	Yes	Yes	Yes

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A reports outcomes for renters in the main sample, Panel B reports outcomes for homeowners in the main sample. Race controls are included in each regression. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7: Effect of TVA relocation on socioeconomic outcomes, county matched sample

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. All Counties</u>				
Treat*Post	0.021* (0.011)	0.028** (0.011)	0.055*** (0.014)	−0.027*** (0.010)
Observations	418,072	418,072	418,072	418,072
R-squared	0.022	0.039	0.026	0.080
Mean	0.499	0.494	0.177	0.159
<u>B. Exclude Counties with Dams Built in the 1930s</u>				
Treat*Post	0.015 (0.010)	0.023** (0.010)	0.056*** (0.013)	−0.034*** (0.010)
Observations	375,814	375,814	375,814	375,814
R-squared	0.025	0.040	0.028	0.077
Mean	0.498	0.497	0.169	0.172
<u>C. Exclude Counties with Dams Built in the 1940s</u>				
Treat*Post	0.020* (0.011)	0.027** (0.011)	0.058*** (0.014)	−0.022** (0.009)
Observations	423,835	423,835	423,835	423,835
R-squared	0.022	0.039	0.027	0.076
Mean	0.494	0.491	0.179	0.162

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A includes the full county matched sample. Panel B excludes counties where dams were built in the 1930s, and Panel C excludes counties where dams were built in the 1940s. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8: Effect of TVA relocation on socioeconomic outcomes, race matched sample

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. White</u>				
Treat*Post	0.010 (0.011)	0.016 (0.012)	0.003 (0.016)	-0.027** (0.012)
Observations	28,181	28,181	28,181	28,181
R-squared	0.029	0.047	0.025	0.054
Mean	0.397	0.382	0.129	0.0626
<u>B. Black</u>				
Treat*Post	-0.028 (0.025)	0.021 (0.021)	0.030 (0.029)	-0.078*** (0.015)
Observations	2,917	2,917	2,917	2,917
R-squared	0.043	0.060	0.062	0.068
Mean	0.399	0.381	0.194	0.0154
Coefficients Different By Race	No	No	No	Yes

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A includes White individuals in the treatment group matched to White individuals in the control group. Panel B includes Black individuals in the treatment group matched to Black individuals in the control group. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

7 Appendix

Figure A1: Stages of dam construction mentioned by Howard (1936)

- ▶ Step 1: Both engineers and social scientists must make numerous investigations. The engineers want to find a firm, physical foundation for their structure and for the finished product to withstand the forces of nature. Before the project is approved or the site is determined, the social scientists need to figure out if the anticipated benefits outweigh the monetary and social costs of the enterprise. If so, they should proceed with the dam construction.
- ▶ Step 2: After the decision to build the dam has been made, there should be focus on questions relating to construction. What labor supply is available within a reasonable distance of the project? What is the composition of the population? What skills are represented by the population? What race factors are present among the population?
- ▶ Step 3: Consideration of the need for a camp to house employees after they are hired which includes awareness of all social implications involved in housing workers in construction communities.
- ▶ Step 4: Understanding the people who are to be employed. What skills are needed? What type of employee training will be most useful? Will the training be on the construction site or will it be held at workers homes?
- ▶ Step 5: Studies are made to assist in planning the educational facilities needed by the employees of the Authority.
- ▶ Step 6: As the actual construction begins, there are a new series of questions to be considered. What commercial facilities must be provided? Are there any desirable recreational facilities? What other economic and social needs must be anticipated?
- ▶ Step 7: Determine the type of community organization that will be most effective such as the community making decisions within popular participation.
- ▶ Step 8: Figure out what cooperative arrangement can be worked out that will be mutually satisfactory to the Authority and the local governments. An example is to consider a reexamination of health facilities due to new demands.
- ▶ Step 9: Social scientists must determine detailed facts concerning families to be displaced by the reservoir.
- ▶ Step 10: As reservoir land is acquired and the people move out of the region, more questions for the public administration and economist. What will be the financial status of the local government after a large portion of an area has been purchased? What source of revenue is left? What is the debt and can it be paid? What is the future of this particular part of the local government?
- ▶ Step 11: As the construction project nears completion, new problems arise. What vocational opportunities are open to workers whose period of employment is about to end? Can the Authority assist them in their efforts to utilize the experience and training received during this period of employment?
- ▶ Step 12: Finally, important adjustments must be made after the construction project is over and people who were involved begin to leave the region.

Table A1: Summary statistics, treatment and control groups

	TG	CG	Difference
<i>Dependent Variables</i>			
Employment	0.397 [0.489]	0.508 [0.500]	-0.112*** (0.011)
Labor Force	0.409 [0.492]	0.542 [0.498]	-0.134*** (0.015)
Unskilled Jobs	0.135 [0.341]	0.187 [0.390]	-0.053*** (0.011)
Skilled Jobs	0.082 [0.275]	0.222 [0.415]	-0.139*** (0.048)
<i>Independent Variables</i>			
White	0.906 [0.292]	0.952 [0.213]	-0.046* (0.026)
Black	0.094 [0.292]	0.048 [0.213]	0.046* (0.026)
Observations	7,924	150,674	

TG refers to individuals in treated counties and CG refers to individuals in control counties. Means are reported for each group with standard deviations in brackets. The Difference column reports coefficients from OLS regressions of each outcome on treatment. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Unskilled Occupations Breakdown, Overall and by Race

VARIABLES	(1) Unskilled (Overall)	(2) Unskilled Without Farming	(3) Unskilled Farming
<u>A. Overall</u>			
Treat*Post	0.064*** (0.015)	0.014** (0.006)	0.009** (0.004)
Observations	317,790	317,790	317,790
R-squared	0.021	0.031	0.040
Mean	0.119	0.064	0.150
<u>B. Black</u>			
Treat*Post	0.120*** (0.025)	0.009 (0.016)	0.007 (0.007)
Observations	16,454	16,454	16,454
R-squared	0.028	0.051	0.039
Mean	0.290	0.214	0.137
<u>C. White</u>			
Treat*Post	0.058*** (0.015)	0.012** (0.006)	0.009* (0.005)
Observations	301,251	301,251	301,251
R-squared	0.008	0.011	0.040
Mean	0.110	0.056	0.151
Coefficients Different By Race	Yes	No	No

Unskilled (Overall) is from our main results and is here for comparison. Unskilled Without Farming considers our unskilled occupations from our main results, but codes any unskilled farming occupations as zero. Unskilled Farming codes any related unskilled farming occupation as one and the rest as zeros. Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A is the overall estimates for each classification of unskilled occupations. Panel B estimates each classification of unskilled occupations for Black individuals while Panel C does so for White individuals. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A3: Effect of TVA relocation on socioeconomic outcomes, county matched sample by race

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. White</u>				
Treat*Post	0.018* (0.011)	0.025** (0.011)	0.052*** (0.015)	-0.031*** (0.011)
Observations	374,775	374,775	374,775	374,775
R-squared	0.023	0.039	0.012	0.085
Mean	0.500	0.494	0.163	0.166
<u>B. Black</u>				
Treat*Post	0.037* (0.022)	0.049** (0.018)	0.090*** (0.024)	-0.004 (0.012)
Observations	43,167	43,167	43,167	43,167
R-squared	0.015	0.036	0.041	0.067
Mean	0.486	0.491	0.299	0.103
Coefficients Different By Race	No	No	No	No

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A reports outcomes for White individuals in the main sample, Panel B reports outcomes for Black individuals in the main sample. Both panels include the full county matched sample. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$